

3rd July

Multifidelity Field Prediction

Problem

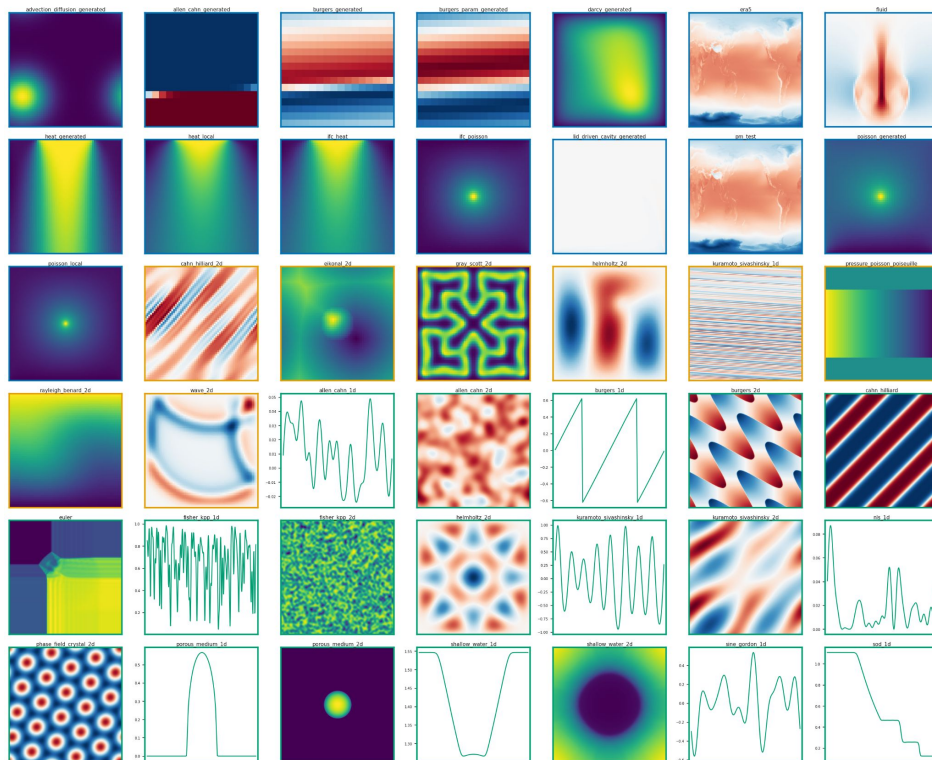
- MF field learning
- HF data is expensive
- There are many methods and no broad, fair, convergence-checked comparison.

Datasets

- 42 datasets
- Spanning
smooth $\leftrightarrow$ sharp
- 1-D $\leftrightarrow$ 2-D
- Fidelity ladders 2 $\rightarrow$ 9

dataset	dim	#fid	HF grid	N_tr	N_te	d	HF $\leftrightarrow$ LF	hi-freq	flag
advection_diffusion_generated	2D	2	64x64	400	100	2	1.00	0.00	
allen_cahn_generated	2D	3	16x16	400	100	2	0.94	0.04	
burgers_generated	2D	3	16x16	400	100	8	0.95	0.00	op-hard
burgers_param_generated	2D	3	16x16	400	100	5	0.93	0.02	op-hard
darcy_generated	2D	3	128x128	400	100	16	1.00	0.00	
era5	2D	9	721x1440	per-fid 1222->65	per-fid 305->7	12	0.99	0.00	
fluid	2D	2	64x64	256	256	2	0.97	0.00	
heat_generated	2D	3	64x64	400	100	3	0.99	0.00	
heat_local	2D	5	128x128	1024	512	3	0.99	0.00	
ifc_heat	2D	4	64x64	100/50/20/5 per fid	128 (HF only)	3	0.94	0.00	op-hard
ifc_poisson	2D	4	64x64	100/50/20/5 per fid	128 (HF only)	5	0.83	0.00	
lid_driven_cavity_generated	2D	3	256x256	400	100	1	0.80	0.09	
pm_test	2D	9	721x1440	same as era5	same as era5	12	0.99	0.00	
poisson_generated	2D	3	64x64	400	100	5	0.98	0.00	
poisson_local	2D	5	128x128	64	512	5	0.98	0.00	
cahn_hilliard_2d	2D	2	64x64	400	100	2	0.33	0.92	
eikonal_2d	2D	2	64x64	400	100	3	0.99	0.00	
gray_scott_2d	2D	2	72x72	400	100	2	0.01	0.00	MF-useless
helmholtz_2d	2D	2	96x96	400	100	3	0.98	0.00	
kuramoto_sivashinsky_1d	2D	2	256x120	400	100	3	-0.06	0.13	MF-useless,op-hard
pressure_poisson_poiseuille	2D	4	64x64	400	100	2	0.93	0.02	
rayleigh_benard_2d	2D	2	64x64	400	100	1	1.00	0.02	
wave_2d	2D	2	80x80	400	100	3	0.85	0.00	
allen_cahn_1d	1D	3	512	400	100	3	1.00	0.00	op-hard
allen_cahn_2d	2D	3	256x256	400	100	3	0.98	0.00	
burgers_1d	1D	3	128	400	100	2	0.94	0.02	
burgers_2d	2D	3	256x256	400	100	2	0.98	0.00	
cahn_hilliard	2D	3	256x256	400	100	19	0.99	0.00	
euler	2D	3	128x128	400	100	7	0.97	0.00	
fisher_kpp_1d	1D	3	512	400	100	2	1.00	0.40	op-hard
fisher_kpp_2d	2D	3	256x256	400	100	2	0.76	0.00	
helmholtz_2d	2D	3	256x256	400	100	2	0.99	0.00	
kuramoto_sivashinsky_1d	1D	3	512	400	100	17	1.00	0.00	op-hard
kuramoto_sivashinsky_2d	2D	3	256x256	400	100	17	0.99	0.00	op-hard
nls_1d	1D	3	512	400	100	17	1.00	0.00	op-hard
phase_field_crystal_2d	2D	3	128x128	400	100	2	0.93	0.00	
porous_medium_1d	1D	3	128	400	100	2	0.98	0.00	
porous_medium_2d	2D	3	256x256	400	100	2	0.99	0.00	
shallow_water_1d	1D	3	128	400	100	2	0.72	0.00	
shallow_water_2d	2D	3	128x128	400	100	2	0.94	0.00	
sine_gordon_1d	1D	3	512	400	100	17	1.00	0.00	op-hard
sod_1d	1D	3	128	400	100	3	0.99	0.01	

Dataset Visualization

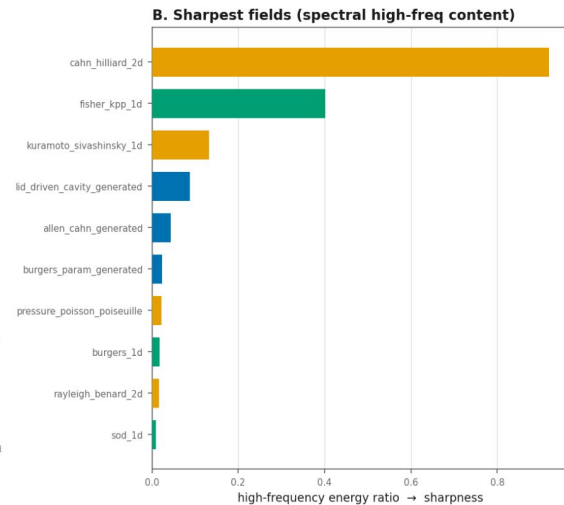
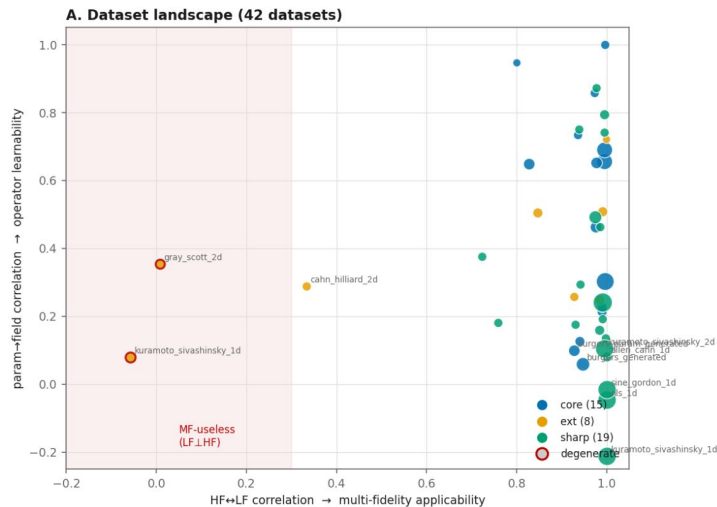


Dataset Science

Every dataset scored on two a-priori axes:

- HF↔LF correlation — is multi-fidelity even valid here? (does cheap data predict expensive?)
- operator learnability — do similar inputs give similar outputs?
- plus sharpness (Fourier high-frequency energy).

Multi-fidelity benchmark — 42 datasets (core · extension · sharp)



9 methods including 5 SOTA Baselines

Model	MF mechanism	Backbone	Conditioning
mf_fno_transfer (Lyu'23)	transfer	FNO	concat
mf_fno_transfer_bar	transfer	FNO	concat (<i>control</i>)
mf_fno_transfer_film	transfer	FNO	FiLM
mf_fno_allpairs	all-pairs ladder	FNO	FiLM + fidelity tag
fno_fire_distcond	distribution-cond.	FNO(ensemble)	LF-uncertainty
fno_coregionalization (IFC)	coregionalization	FNO	cross-fidelity latent
mfrnp (Niu'24)	residual process	Neural Process	context
mf_deeponet	LF+correction	DeepONet	branch(params)
transolver_residual (Wu'24)	residual	Transformer	attention tokens

Conditioning wins — FiLM is #1

#	model	ELO	geomean rel-L2
1	mf_fno_transfer_film	1725	0.017
2	mf_fno_allpairs	1664	0.018
3	fno_fire_distcond	1566	0.029
4	mf_fno_transfer_bar (concat)	1566	0.021
5	mf_fno_transfer (concat)	1543	0.021
6	transolver	1511	0.033
7	fno_coregionalization	1392	0.078
8	mfrnp	1316	0.225
9	mf_deeponet	1218	0.153

Error Summary

- Difficulty is intrinsic to the field, not the fidelity relationship:
 - rough fields (TV, $p=+0.56$) and chaotic operators (learnability, $p=-0.64$) are hard for every model — the same dark columns across the whole heatmap.
- HF $\leftrightarrow$ LF correlation does *not* predict error ($p\approx 0$): it governs whether MF is *valid* (Fig 1), not task difficulty.

